

Onboarding, Multimodal Data Collection, and Waveform Optimization for Radar-Based Human Activity Monitoring

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Abstract—This paper summarizes my work over the quarter on onboarding to a multimodal radar sensing system and optimizing an FMCW waveform for healthcare-oriented human activity monitoring. A portion of the quarter was spent in the Living Lab, the off-campus lab in the Medicine Building, where I learned to operate the radar system and participated as the subject in several motion scenarios, including sit-to-stand, freezing and shuffling gait, fall events, and long mixed-activity sequences. These captures were intended to simulate sensing in a hospital-room-like environment and helped establish practical familiarity with the system.

In parallel, I evaluated a new indoor-oriented radar waveform against an older waveform that had been tuned for outdoor use. The comparison focused on sweep bandwidth, range and velocity resolution, duty cycle, and frame rate, along with corner reflector testing to verify detectability and range behavior. Overall, the work helped me become a more independent contributor to the lab while also producing a waveform configuration better aligned with close-range indoor sensing.

I. INTRODUCTION

Radar-based human activity monitoring is a promising approach for healthcare applications because it can capture motion patterns without requiring direct contact or visible imaging. In our lab, the broader objective is to study human motion for clinically relevant tasks such as Parkinson’s disease assessment, gait monitoring, and activities of daily living. My work this quarter focused on two connected parts of that effort: becoming independently comfortable with the radar capture workflow and improving the waveform used for indoor sensing.

A portion of the quarter was spent at the Living Lab, the off-campus lab in the Medicine Building. That period provided hands-on experience with the radar system in a more realistic indoor setting, although it was only one part of the full quarter. During those sessions, I participated in captures designed to simulate a hospital-room environment. I helped run captures and also served as the subject for motions such as sit-to-stand, freezing or shuffling gait, fall events, straight walking, turning, and long combinations of activities. That experience improved my understanding of how the radar behaves in realistic indoor settings and what kinds of motions are most important to capture clearly.

In the second part of the quarter, I worked on waveform optimization. The new indoor waveform was designed for indoor use and emphasizes higher temporal sampling and

a higher duty cycle. I compared it against the old outdoor waveform using corner reflector tests and motion scenarios to understand the tradeoffs between the two settings.

II. SYSTEM OVERVIEW

The sensing system is built around a ROS2-based capture pipeline that supports multimodal recording and offline analysis. The radar data is logged during capture sessions and later inspected in Python for range, Doppler, and motion analysis. The broader system also includes Bosch UWB integration, although that was not the main focus of my work this quarter.

The radar platform is a TI xWR68xx FMCW system. The waveform parameters control the range coverage, velocity sensitivity, frame timing, and overall suitability for the target environment. In this project, the main practical distinction was between a waveform designed for higher-tempo indoor sensing and a waveform originally used for a more outdoor-oriented setup.

The main application context is healthcare motion understanding. The captures are intended to support analysis of gait, freezing episodes, sit-to-stand transitions, falls, and multi-person interactions. Because the goal is to understand motion in a hospital-like room, the waveform and capture setup need to balance range coverage, temporal resolution, and robustness to indoor clutter.

III. METHODOLOGY

My work this quarter had two main components: system onboarding and waveform evaluation. For onboarding, I began by learning how the radar capture flow works, then moved into guided sessions, and eventually participated in captures with limited supervision. This process helped me transition from learning the workflow to independently supporting meaningful capture sessions.

For the Living Lab portion, I participated in off-campus capture sessions in the Medicine Building. During these sessions, I also served as the subject in the capture environment, which is shown in Fig. 1.



Fig. 1: Living Lab capture session in the Medicine Building. The screenshot shows the capture environment during one of the motion sessions used to simulate a hospital-room setting.

This photo provides a simple visual reference for the type of hands-on data collection performed during the Living Lab portion of the quarter. The goal was to collect data in a setting that more closely resembles a hospital room than the main lab. I contributed by helping with setup and capture flow and by serving as the subject during several activity sequences. The activities included sit-to-stand, freezing and shuffling gait, fall events, straight walking, turning, and long mixed-activity recordings.

For the waveform work, I compared the new indoor waveform against the old outdoor waveform using both parameter analysis and experimental testing. The parameter comparison focused on bandwidth, duty cycle, frame rate, and resolution tradeoffs. I then used corner reflector tests to verify that the new waveform could still detect a strong target across the intended range of distances. I also used multi-person and gait-based motion scenarios to assess whether the new waveform preserved useful structure in Range-Doppler and Range-Azimuth tests.

IV. QUARTER PLAN AND DEVIATION

At the start of the quarter, the main plan was to get me onboarded to the sensing system, transfer the necessary skills, and move me toward independent operation. The original plan also included supporting UWB integration and contributing to waveform or system improvements. In practice, I made solid progress on onboarding and capture operations, but a larger share of the quarter shifted toward Living Lab data collection and waveform optimization.

That shift was not a problem, since the work still aligned with the lab's larger goals. The Living Lab sessions provided the hands-on experience needed to understand the system in a realistic indoor setting, and the waveform work produced a concrete technical improvement. The main task that remained incomplete was Bosch UWB integration, which received less time once waveform evaluation became a larger focus. One practical issue that came up was that the LiDAR stopped working right before the start of the quarter, so captures did not include LiDAR data.

Overall, the quarter was less about strictly following the original task structure and more about establishing a stronger foundation for future work.

V. WAVEFORM CONFIGURATION AND IMPROVEMENTS

A. Waveform Design Rationale

I designed the new indoor waveform to improve close-range indoor sensing by increasing the sweep bandwidth while keeping the chirp timing practical for the system. For an FMCW waveform, the range resolution is approximately

$$\Delta R = \frac{c}{2B}, \quad (1)$$

where c is the speed of light and B is the sweep bandwidth. Since range resolution improves as bandwidth increases, one of the main objectives of the redesign was to push the bandwidth toward 4 GHz while maintaining enough resolution to separate nearby targets in indoor scenes.

For an FMCW chirp, the effective sweep bandwidth can be related to the chirp slope and the ADC sampling window by

$$B = ST_{\text{adc}}, \quad (2)$$

where S is the chirp slope and T_{adc} is the ADC sampling time. The ADC window is given by

$$T_{\text{adc}} = \frac{N_s}{F_s}, \quad (3)$$

where N_s is the number of ADC samples per chirp and F_s is the sampling rate. For the new indoor waveform, this corresponds to $N_s = 256$ and $F_s = 3250$, so the sampled chirp duration is fixed by those settings. To increase bandwidth, I increased the chirp slope while keeping the sampled window large enough to preserve fine range detail. At the same time, I reduced the idle time between chirps so that a larger fraction of the frame period was devoted to useful sensing rather than waiting between sweeps.

The effective duty cycle can be written as

$$D = \frac{T_{\text{active}}}{T_{\text{frame}}} \times 100\%, \quad (4)$$

where T_{active} is the total active sensing time in a frame and T_{frame} is the frame period. By reducing idle time and increasing the portion of the frame spent on chirps, the new indoor waveform achieved a much higher duty cycle than the old outdoor waveform. In practice, this means the radar spends more time sensing and less time idle, which is useful for tracking continuous human motion.

The resulting tradeoff is a reduction in maximum range, but that tradeoff is acceptable for this project because the intended environment is a hospital-room-like indoor setting, where close-range motion detail is more important than long-range coverage.

Overall, the new indoor waveform was designed to favor finer range resolution and more continuous sampling. This makes it better suited to indoor human activity monitoring,

where the primary objective is to resolve motion clearly within a limited sensing region.

Table I summarizes the parameters of the new indoor waveform, the indoor-oriented waveform used in this work.

TABLE I: New indoor waveform configuration summary

Parameter	Value
Starting Frequency (GHz)	60
Slope (MHz/ μ s)	50.6
Samples per chirp	256
Chirps per frame	238
Sampling rate (Mps)	3.25
Sweep Bandwidth (GHz)	3.986
Frame periodicity (ms)	100 (10 fps)
Transmit antennas	3
Receive antennas	4
Chirp Ramp time (μ s)	79
Chirp Idle Time (μ s)	60
Duty Cycle	99.25%
Maximum unambiguous range (m)	9.63
Maximum radial velocity (m/s)	3.00
Azimuth resolution (deg)	14
Range resolution (m)	0.0376
Radial velocity resolution (m/s)	0.094
Elevation resolution (deg)	57

Table II summarizes the main differences between the new indoor waveform and the old outdoor waveform.

TABLE II: Key improvements of the new indoor waveform relative to the old outdoor waveform

Parameter	Old	New	Change
Sweep bandwidth	2.0 GHz	3.986 GHz	+99%
Range resolution	0.075 m	0.0376 m	50% better
Max range	19.13 m	9.63 m	Reduced tradeoff
Duty cycle	60.05%	99.25%	65% increase
Velocity resolution	0.188 m/s	0.094 m/s	50% better
Frame rate	45 fps	10 fps	Slower tradeoff

The main improvement is that the new indoor waveform is much more continuous in time and gives better range and velocity resolution for close-range indoor sensing. The main tradeoff is that it reduces maximum range and lowers the frame rate, which is acceptable for the intended hospital-room use case.

VI. RESULTS

A. Corner Reflector Testing

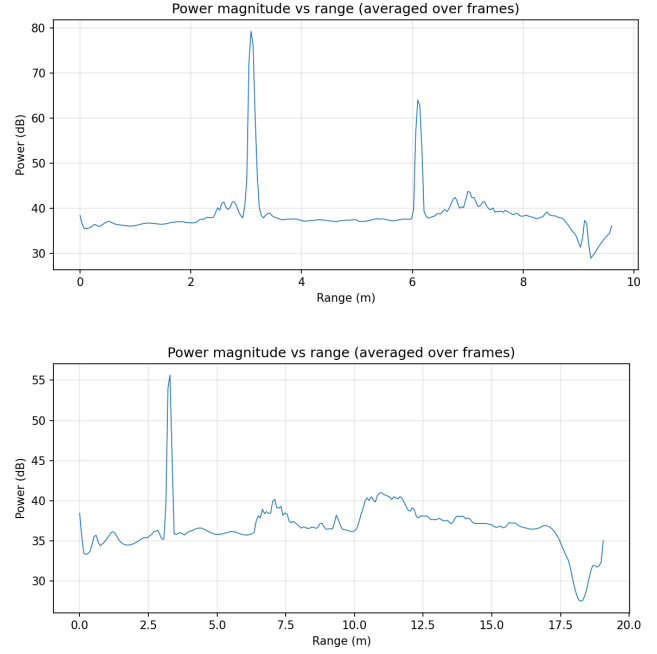


Fig. 2: New indoor waveform (top) and old outdoor waveform (bottom) at 3 m.

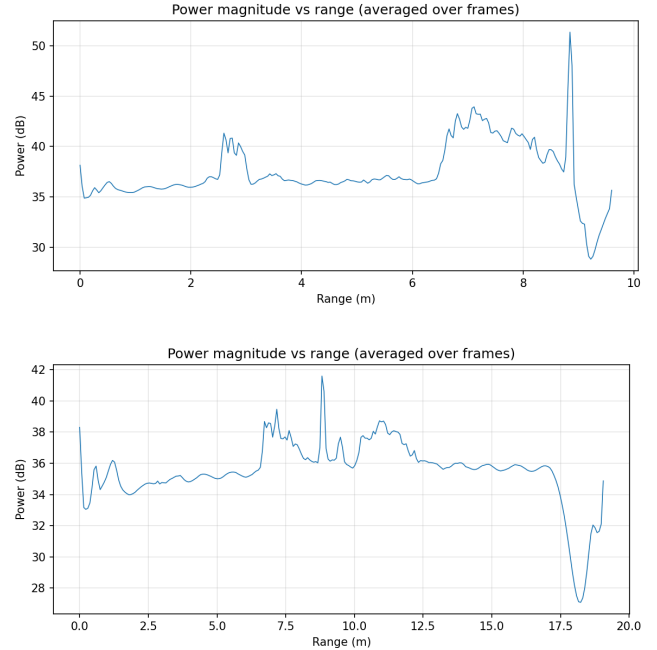


Fig. 3: New indoor waveform (top) and old outdoor waveform (bottom) at 9 m.

I tested corner reflector captures at 3 m and 9 m to compare the new indoor waveform with the old outdoor waveform.

For each distance, the new indoor waveform is shown on top and the old outdoor waveform is shown below. The plots are highly similar, which indicates that the new indoor waveform preserves reliable reflector detectability while shifting the sensing tradeoffs toward a more indoor-friendly operating point.

B. Micro-Doppler Motion Capture Comparison

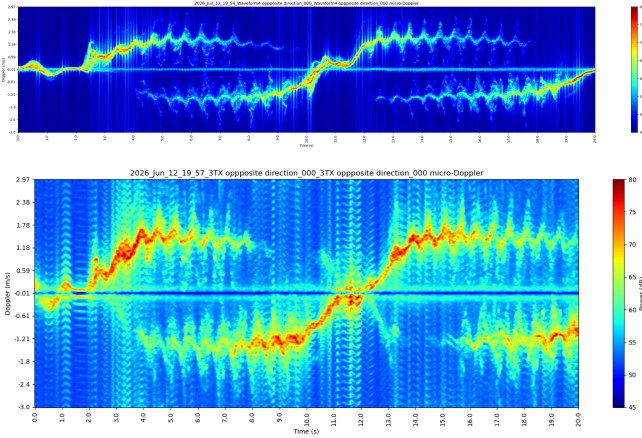


Fig. 4: uD comparison for two-person walking in opposite directions: new indoor waveform (top) and old outdoor waveform (bottom).

Figure 3 shows the micro-Doppler spectrograms for the two-person opposite-direction walking scenario captured with the new indoor waveform and the baseline waveform. Both spectrograms span the same 20-second interval and cover a velocity range of ± 3.0 m/s. The scenario consists of two subjects walking toward each other, passing one another, continuing in opposite directions, turning around, and finally walking back towards each other, which produces simultaneous Doppler signatures of opposite sign.

The most visible difference between the two waveforms is the noise floor. The new indoor waveform produces a noticeably darker background and higher contrast between the walking signatures and clutter, while the baseline waveform smears the motion components and the background. This improvement is consistent with the higher duty cycle and larger coherent observation window of the new indoor waveform, which supports stronger Doppler integration and better suppression of incoherent noise.

C. Sky Test

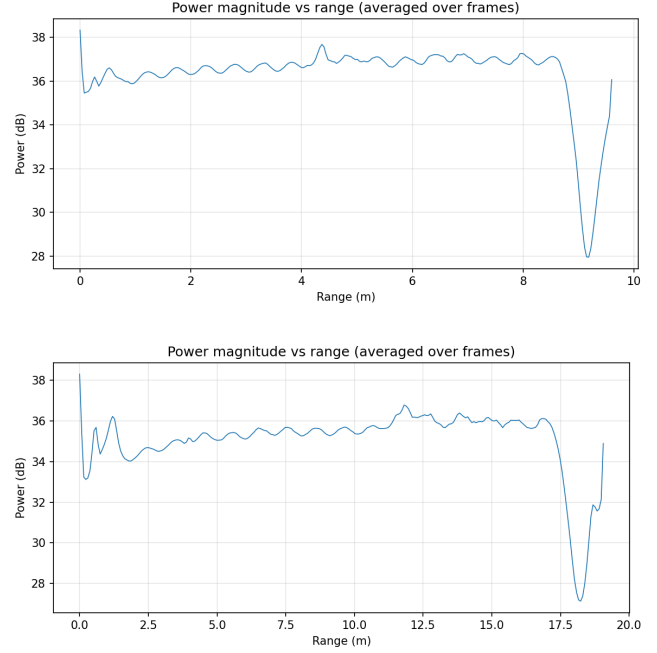


Fig. 5: Sky test range-profile comparison: new indoor waveform (top) and old outdoor waveform (bottom).

D. Sky Test

To check the baseline behavior of the radar system, I conducted a sky test with no intentional target present. In this case, the range profile should remain mostly smooth, since any strong peak would likely come from leakage, coupling, or another internal effect rather than a real target.

The measured range profiles are generally smooth for both waveforms, which suggests that neither one produces strong false targets under these conditions. Both profiles show a noticeable rise near zero range, which is expected and usually comes from direct coupling between the transmit and receive paths. Away from this near-range region, the remaining variation is small and does not show clear mid-range clutter. Both profiles also taper off near the far end of the plotted range, which is consistent with weaker signal levels at the limits of the display.

Because I did not have a Doppler-spread plot for this test, I treated it mainly as a basic range-profile check. Even so, the lack of clear peaks in the middle of the range supports the conclusion that both waveforms behave in a stable way when no target is present.

E. Summary of Observations

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Across the experiments in this quarter, the new indoor waveform showed the clearest advantage in close-range sensing and motion visibility. The corner reflector tests showed that it still produced reliable returns at the tested distances, while the motion capture comparison test showed cleaner micro-Doppler

structure and a lower background level. That motion test also supported the benefit of improved range resolution, since the new waveform is better suited to separating nearby motion components when two subjects are close to one another. The sky test also suggested that the waveform behaves stably when no target is present.

The main tradeoff is reduced maximum range and a lower frame rate compared with the older outdoor waveform. For the intended indoor and hospital-room-like setting, however, that tradeoff is acceptable because the priority is not long-range coverage, but clearer motion detail and better separation of nearby activity.

VII. FUTURE WORK AND CONCLUSION

Future work will focus on integrating the sensing stack more fully across radar, Bosch UWB, and other available modalities. During the quarter, I was present for a UWB capture test of the board as well as a Bosch meeting with Sushanta, Edison, and Shubo, which helped clarify how the UWB work fits into the broader system. The next step is to continue bringing these systems together in a way that supports repeatable data collection without requiring extensive manual setup each time.

A longer-term goal is to make the sensing pipeline much easier to deploy and use, so that the full collection process can eventually be packaged into a more self-contained workflow. In that vision, the sensors would be integrated into a more portable and easier-to-run setup, allowing other users to deploy the system and collect large-scale datasets with less friction. This is especially relevant for eventual hospital use, where the goal is to capture real motion data in clinical environments rather than only in lab settings.

In conclusion, this quarter established a stronger foundation for independent operation of the radar system while also advancing waveform design and system understanding. The work on Living Lab captures, waveform optimization, and early exposure to Bosch UWB and LiDAR-related integration all contributed to a clearer path toward a more complete multimodal sensing platform.

REFERENCES

- [1] TI mmWave SDK Documentation.
- [2] FMCW radar tutorial materials used during onboarding.
- [3] S. Hor, S. Yang, J. Choi, and A. Arbabian, "MVDoppler: Unleashing the Power of Multi-View Doppler for MicroMotion-Based Gait Classification," Stanford University. [Online]. Available: <https://mvdoppler.github.io>